Oskar Skorge 20000414-4515

Advanced Analysis of Economic Change

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Intergenerational Effects of Wealth and Income: A Critical Review of Toney and Robertson (2021)

A central question in the study of economics concerns inequality of opportunity and initial conditions that shape the income-generating opportunities of future generations. Indeed, the intergenerational persistence and crystallization of wealth warrant further investigation into the mechanisms that shape opportunities and catalyse social mobility. Toney & Robertson contribute to this debate by examining parental wealth effects on their children's wealth and income-generating levels using the Panel Study of Income Dynamics (PSID) in the US, spanning 1984-2018. The paper's identification strategy follows a 2SLS IV design in which parental wealth is assumed to be an exogenous instrument for child stock holdings, thereby raising income levels. The study suggests that a one percent increase in stock value through this channel raises estimated concurrent household income by 0.18-0.25 percent.

The first stage estimation thereby estimates the effect of the extended family wealth on younger generation stock holding, $LnStocks_{it}$ using C'_{it} a vector representing child characteristics, parental wealth, and financial behaviour variables (see Appendix Table 8).

$$LnStocks_{it} = C'_{it}\alpha + u_{it}$$

For the second stage, the causal effect of stock value is associated with log household income, where $LnInc_{it}$ is the IHS-transformed household income for adult child i in survey year t X'_{it} contains age, age squared, and race:

$$LnInc_{it} = X'_{it}\beta + \text{LnStocks}'_{it}\phi + \varepsilon_{it}$$

While the study seeks to ascertain the role of intergenerational wealth as a significant enabler of children's future income-generating effects, we propose that the study's exclusion criteria are not met. Parental wealth may have both direct and indirect effects on children's income-generating outcomes beyond stock holdings. As such, this replication exercise seeks to proceed in three parts. First, we reproduce the core empirical results: the 2SLS estimates, the first-stage regressions, and the Oaxaca-Blinder decompositions of ethnic differences. Second, we evaluate the identification strategy, focusing on the exclusion restriction by conducting a Sargan/Hansen J test and a Wu-Hausman test to assess potential overidentification and endogeneity. Finally, we extend the analysis by re-specifying the instrumental-variables design with a more defensible partition of controls and instruments of personal endowments in the second stage of the 2SLS design to test the causal claims, effect sizes, and the validity of the methodology in its pursuit to identify the role of wealth and intergenerational economic mobility across ethnicities.

1. Replication

1.1.The relationship between intergenerational wealth and income

The first stage results suggest that parental wealth is a significant predictor of child stock value (Table 1). While our paper has replicated the overall findings, there are discrepancies in the effect sizes (Table 2). This may be due in part to the original paper utilising random-effects EC2SLS estimators, which account for panel standard errors (Baltagi, 1981). Our replication in R Studio uses pooled 2SLS with cluster-robust standard errors. Despite this, we still observe a positive relationship between parental wealth and, to some extent, maternal grandparents' wealth and children's stock value, indicating potential intergenerational persistence of wealth.

Table 1. First Stage Results of Younger Generation's Stock Value Influenced by Parental and Grandparental Wealth

	Stock Value (1)	Stock Value (2) Paternal GP	Stock Value (3) Maternal GP
Parent's Wealth	0.029***		
	(0.007)		
Grandparent's Wealth		0.004	0.072*
		(0.014)	(0.040)
Person-Year Observations	12,835	3,578	458
R-squared	0.1617	0.1865	0.1964
Number of Children	1,948	1,032	458

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Standard errors in parentheses. Each column reports the first stage results from 2SLS/IV panel regressions. Dependent variable: IHS log of stocks and mutual funds held outside pension wealth. Dependent variable: IHS log of stocks and mutual funds held outside pension wealth. Note: The replication data package provides only the 2017 cross-section for the maternal grandparent–grandchild sample (N = 1,514), rather than the full panel (N = 3,774) reported in the original paper.

Table 2. Replication Comparison: Wealth Effect Family – Child Relation

Sample	Variable	Original Coef.	Original SE	Original R-sq	Replicated Coef.	Replicated SE	Replicated R-sq
Parent-Child	Parent's Wealth	0.0293	0.0066	0.1617	0.0293	0.0066	0.1617
Paternal GP-GC	Paternal GP Wealth	0.0029	0.0143	0.1881	0.0037	0.0143	0.1865
Maternal GP-GC	Maternal GP Wealth	0.0463	0.0136	0.1950	0.0716	0.0397	0.1964

Note: Minor discrepancies may occur due to differences between Stata's xtivreg random-effects EC2SLS estimator and R's pooled 2SLS approximation with ivreg. The Stata files in Toney and Robinson (2008) feature a random-effects error structure, while this R implementation employs pooled 2SLS with cluster-robust standard errors.

For the second-stage estimates (Table 3), the instrument channel across all three generational linkages remains positive and highly significant, despite minor discrepancies in the estimated effects (Table 4). The income elasticity remains largest for parent-child pairs, consistent with the hypothesis of Toney & Robertson (2021) that wealth effects attenuate, yet remain sticky across generations, pointing to broader implications of social mobility beyond familiar background.

Table 3. 2SLS/IV Coefficients of Stock Value Influenced by Parental and Grandparental Wealth on the Younger Generation's Income

	Parent-Child	Paternal GP-GC	Maternal GP-GC
Log stock value (instrumented)	0.306***	0.259***	0.229***
	(0.016)	(0.024)	(0.046)
Age	0.092***	0.080***	0.514**
	(0.018)	(0.026)	(0.252)
Age squared	−0.001***	−0.001***	−0.006*
	(0.000)	(0.000)	(0.003)
Race (Black = 1)	−0.084*	−0.131*	−0.125
	(0.045)	(0.072)	(0.151)
Constant	9.489***	9.803***	0.619
	(0.327)	(0.474)	(5.247)
Person-Year Observations	12,835	3,578	458
R-squared	−0.606	−0.397	−0.117
Number of Children	1,948	1,032	458

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Cluster-robust standard errors in parentheses. Note: Each column reports the second-stage results from 2SLS/IV panel regressions. Log of stock values is instrumented with (grand)parental wealth, income, education, demographics, and time dummies. Dependent variable: IHS-transformed household income. The maternal sample is substantially smaller due to missing data on risk preference and wealth, reducing the number of observations from 1514 to 458.

Table 4. Replication Comparison: Stock Value Coefficient on Household Income

Sample	Original Coef.	Original SE	Replicated Coef.	Replicated SE	Diff.
Parent-Child	0.2523	0.0133	0.3063	0.0163	0.0540
Paternal GP-GC	0.2200	0.0191	0.2589	0.0237	0.0389
Maternal GP-GC	0.1826	0.0130	0.2290	0.0460	0.0464

Note: Minor discrepancies may occur due to differences between Stata's xtivreg random-effects EC2SLS estimator and R's pooled 2SLS approximation with ivreg. The Stata files in Toney and Robinson (2008) feature a random-effects error structure, while this R implementation employs pooled 2SLS with cluster-robust standard errors.

1.2.De-Composition of Endowments by Ethnicity

We complement the IV analysis with an Oaxaca-Blinder decomposition (Table 5) that partitions the Black–White income gap into an endowment component reflecting differences in observable characteristics such as wealth and a coefficient component, capturing differential returns to those characteristics across racial groups. The findings suggest that observable family characteristics, rather than differences in returns to the same characteristics, remain a prevalent driver of income inequality between Black and White households in the US (Figure 1).

Table 5: Oaxaca-Blinder Decomposition of the Black-White Income Gap (2017)

	(1) Parent-Child	(2) Paternal GP-GC	(3) Maternal GP-GC
Differential			
White Household Income (log)	12.1473	11.9526	11.9272
Black Household Income (log)	11.4131	11.0706	11.0084
White−Black Income Gap	0.7342	0.8819	0.9188
Decomposition			
Endowments	0.5028***	0.7530***	0.5338***
	(0.0694)	(0.1165)	(0.0924)
Coefficients	0.1738**	0.1493	0.2136***
	(0.0854)	(0.1148)	(0.0712)
Interaction	0.0575	-0.0775	-0.0519
	(0.0807)	(0.1196)	(0.0971)
Observations	1,809	1,405	1,514

Note: Threefold Oaxaca-Blinder decomposition. Standard errors in parentheses (bootstrapped, R = 100). White is the reference group. Dependent variable: IHS household income. Column (1) uses parental income and wealth; columns (2)-(3) use grandparental income and wealth. *** p<0.01, ** p<0.05, * p<0.10

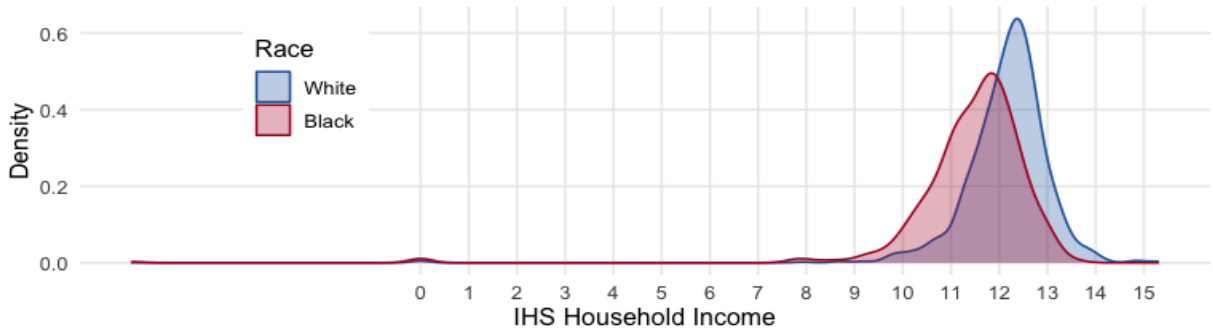

Figure 1. Kernel Density of IHS Household Income by Race, 2017 Cross-Section.

2. Discussion

Although the study of intergenerational wealth and its effects is a considerable motivation for research on social mobility, a prominent issue in the research design of Toney & Robertson (2021) is their identification strategy. For the 2SLS to yield consistent estimates of the stock-income elasticity (φ), every variable in C_{it} the model that is excluded from the second stage must affect household income *only* through stock holdings.

The implication that family wealth only works through stock holdings is a strong and implausible assumption. For one, educational attainment is among the most robust predictors of earnings in labour economics (Mincer, 1974; Card, 1999), operating through human capital channels entirely independent of stock market participation, yet it can be enabled by high-level private education endorsed by wealthy families. This example is not a marginal concern, but points to the first-order criteria of the dependent variable, which are routed exclusively through the first stage. The consequence is that φ captures not only the effect of stock holdings on income but also the direct effects of education, employment, and family structure, inflating the estimated wealth–income elasticity and implying that the exclusion criteria is likely not met.

A deeper conceptual issue concerns the assumed transmission channel. The paper frames stock holdings as the mechanism through which family wealth raises children's income, yet equities held outside pension accounts represent a narrow and skewed slice of household portfolios, disproportionately held by white, higher-income families. Further channels through which intergenerational wealth influences income are housing, educational investment, and social capital (Pfeffer and Killewald, 2018; Chetty et al., 2014), none of which are captured by the stock variable or during the second stage.

In effect, the paper instruments a proxy with variables that are themselves direct determinants of the outcome, producing an estimate that is neither cleanly causal nor interpretable as a wealth effect. Furthermore, employing more than 15 instruments for a single endogenous regressor raises concerns about overidentification bias, which can push 2SLS estimates toward OLS and erode the estimator's properties (Bound et al., 1995). The absence of any reported first-stage F-statistic, Sargan/Hansen J-test, or Wu-Hausman test leaves the reader with no basis for assessing instrument relevance or validity, a significant omission for a paper whose contribution rests on causal identification through IV, which is accounted for and controlled in our extension.

3. Extension

3.1.Instrument and Endogeneity Tests

Under the null hypothesis, all instruments are valid and uncorrelated with the structural error term. A rejection of this null provides formal statistical evidence that the exclusion restriction is violated, that is, at least one instrument has a direct effect on income beyond the stock-holdings channel. Given the theoretical concerns outlined above, a rejection of Ho is expected. While the first-stage results suggest strong instrument relevance, the overidentification test rejects the null of instrument exogeneity, indicating that the exclusion restriction may be violated. As raised in our discussion, the IV estimates should therefore be interpreted with high caution.

Table 6. Weak Instrument, Endogeneity, and Overidentification Tests

Test	Statistic	p-value
Weak Instruments (first-stage F)	104.15	0.0000
Wu-Hausman (endogeneity)	1656.94	0.0000
Sargan (overidentifying restrictions)	931.74	0.0000

Notes: Weak instruments: first-stage F-statistic; rule of thumb F > 10 (Stock and Yogo, 2005). Wu-Hausman: tests endogeneity of the instrumented regressor. Sargan: tests overidentifying restrictions; rejection implies at least one instrument is invalid.

3.2. Second Stage Controls and Endowment Effects

To address the exclusion restriction violation, we re-specify the 2SLS model by moving the six child-level variables: educational attainment, employment status, occupational class, marital status, sex, and number of children from the excluded instrument set to the second stage as controls (Table 7). The corrected specification retains parental wealth, pension ownership, homeownership, computer use, risk avoidance, and year dummies as excluded instruments.

Table 7: Original vs. Corrected IV Specification (Parent-Child Sample) of Stock Value and IHS Income.

	(1) Original	**(2) ¨Second Stage Controls**
Log stock value (instrumented)	0.306***	0.210***
	(0.016)	(0.015)
Age	0.092***	0.046***
	(0.018)	(0.016)
Age squared	−0.001***	−0.001**
	(0.000)	(0.000)
Race (Black = 1)	−0.084*	−0.024
	(0.045)	(0.036)
Educational attainment		0.034***
		(0.010)
Married		0.568***
		(0.041)
Employed		0.422***

		(0.042)
Professional/managerial		−0.100***
		(0.035)
Sex (Male = 1)		0.058
		(0.049)
No. children in HH		0.014
		(0.013)
Constant	9.489***	9.334***
	(0.327)	(0.297)
Person-Year Observations	12,835	12,835
First-stage F	104.2	75.7
Sargan J (p-value)	0.0000	0.0000

Note: Column (1) reproduces the original specification with age, age², and race as second-stage controls. Column (2) moves six child-level variables, educational attainment, marital status, employment status, occupational class, sex, and number of children, from the excluded instrument set to the second stage. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The first-stage F-statistic remains well above the Stock and Yogo (2005) threshold (75.7 > 10), indicating that instrument relevance is preserved despite reduced instrument count. The estimated stock-income elasticity, however, is approximately one-third lower (from 0.306 to 0.210), confirming that the original estimate could be inflated by the direct effects of omitted controls channelled through the instrumented variable. Notably, the race coefficient loses statistical significance. From the original second stage, we suggest race was absorbing the combined effects of racial disparities in education, employment, and marriage, structural inequalities that, once controlled for, leave little residual for the race binary by itself. The Sargan J-test remains significant, suggesting that the exclusion restriction is not satisfied.

4. Conclusion

While our replication exercise has reproduced the original findings of Toney and Robertson (2021) that intergenerational wealth transfers, mediated by financial asset accumulation, are positively associated with children's household income, our extension exposes a material weakness in their identification strategy. The exclusion restriction is violated when direct income determinants are included as instruments, inflating the estimated wealth–income elasticity by roughly one-third. When corrected, the stock coefficient remains significant but substantially attenuated, and the racial income penalty disappears once human capital and labour market characteristics are controlled for in the second stage, reflecting structural inequalities in endowments, consistent with the Oaxaca-Blinder decomposition results, rather than being determined by intergenerational wealth effects through financial markets alone.

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6. Appendix

Table 8. Variable Definitions – Original Paper

Variable	Description	Role
Outcome and Endogenous Variable		
isib_famincome	IHS household income of adult child	Dependent (2nd stage)
isib_stocks	IHS stock and mutual fund value (outside pension)	Endogenous (instrumented)
Family Background		
ipar_wealth	IHS parental wealth	Instrument/Control
ipar_inc	IHS parental income	Control
igpar_wealth	IHS grandparental wealth	Instrument/Control
igpar_inc	IHS grandparental income	Control
Demographics		
isib_age	Age of adult child	Control
sib_agesq	Age squared	Control
isib_race	Race (1 = non-Latino Black, 0 = non-Latino White)	Control
First-Stage Instruments		
isib_educ	Educational attainment	Instrument
isib_child	Number of children in household	Instrument
isib_married	Marital status (1 = married)	Instrument
isib_employ	Employment status	Instrument
isib_occupation	Professional or managerial occupation (1 = yes)	Instrument
sib_pension	Pension ownership	Instrument
isib_2home	Homeownership	Instrument
isib_computeruse	Computer/internet use	Instrument
isib_riskpref	Risk avoidance	Instrument
sib_sex	Sex (1 = male)	Instrument
Weight		
sib_famweight	Family sampling weight	Weight (decomposition)